

HCSM-41

Absence of a Light Dark Photon

The HCSM Prediction $\epsilon = 0$ Exactly, from the Absence of a Hidden Abelian Gauge Factor and the $U(1)_Y$ -Neutrality of the Hidden Sector

Hodge Complex Standard Model

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Abstract

The dark photon A' the lightest Kaluza-Klein gauge mode of a hidden Abelian gauge symmetry is a widely studied candidate for the mediator of dark matter and for a low-mass extension of the Standard Model. In this paper we derive the Hodge Complex Standard Model (HCSM) prediction for the dark photon: it does not exist. The dark photon is absent from the HCSM spectrum because the HCSM gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ (HCSM-06) contains no hidden Abelian factor that couples to electrically charged matter, and the hidden sector content $5B_1 \oplus 6A_1 \oplus 31$ (HCSM-38 [8]) is $U(1)_Y$ -neutral (HCSM-39 [9], Disclosed Choice 2). Consequently:

- No tree-level kinetic mixing operator $F_{\mu\nu}F'^{\mu\nu}$ can be written, because no A'_μ exists to mix.
- No one-loop kinetic mixing is generated, because no state is charged under both $U(1)_Y$ (or $U(1)_{\text{em}}$) and any hidden Abelian factor.

We prove the exact result

$$\epsilon \equiv 0 \text{ over all mass scales.}$$

The prediction is falsifiable by every major dark photon search: SHiP, FASER2, MATHUSLA, CODEX-b, LDMX, NA64, Belle II, and HL-LHC. Any confirmed dark photon with $\epsilon > 10^{-12}$ at any mass below the electroweak scale falsifies the HCSM framework. This paper completes the $\mathfrak{u}(1)_Y$ gauge sector of the HCSM stack by closing the final gauge content prediction.

Derivation status summary. *Derived:* the absence of a hidden Abelian gauge factor coupling to electrically charged matter; the absence of a tree level kinetic mixing operator; the absence of one loop kinetic mixing; the exact prediction $\epsilon \equiv 0$ over all mass scales. *Disclosed inputs:* the gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ (HCSM-06); the hidden sector content $5 + 6 + 31$ (HCSM-38); the $U(1)_Y$ -neutrality of hidden states (HCSM-39, Disclosed Choice 2). *Open:* the precise form of the interaction between the four hidden $\mathfrak{u}(1)$ factors and the hidden sector $5 + 6 + 31$; whether the hidden sector carries a different stabilising symmetry that could serve the role usually assigned to the dark photon (see HCSM-45e in the framework's programme).

Contents

1	Introduction	2
1.1	The dark photon paradigm	2
1.2	The HCSM prediction	3
1.3	On the original content of HCSM-41	3
1.4	Summary of results	3
1.5	Organization	4

2	Setup	4
2.1	The substrate and the gauge algebra	4
2.2	The hidden sector	4
2.3	The hypercharge functional and its neutral kernel	4
2.4	The visible $u(1)_{\text{em}}$	5
3	Absence of a hidden Abelian gauge factor coupling to charged matter	5
3.1	The five Abelian factors	5
4	Absence of tree-level kinetic mixing	6
5	Absence of one-loop kinetic mixing	6
6	The prediction $\epsilon \equiv 0$	7
7	Falsifiability	7
8	Comparison with the A_5 / Randall–Sundrum framework	7
8.1	The A_5 dark-photon prediction	7
8.2	The numerical coincidence and its resolution	8
8.3	Distinguishing test	8
8.4	Implications for the HCSM stack	8
9	Derivation status	8
9.1	Derived from the substrate	8
9.2	Disclosed inputs	8
9.3	Open	9
10	Falsifiable predictions	9
11	Conclusion	9
A	Reproducible verification	10
B	Numerical values	10
C	Web Crawler Metadata	10
C.1	Dublin Core	10
C.2	BibTeX	11

1 Introduction

1.1 The dark photon paradigm

The dark photon A' a hypothetical light gauge boson of a new Abelian symmetry $U(1)'$ — is one of the most widely studied extensions of the Standard Model. It is characterized by a mass $m_{A'}$ in the MeV–GeV range and a small kinetic mixing parameter ϵ with the Standard Model photon, appearing in the effective Lagrangian through

$$\mathcal{L} \supset -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{4}F'_{\mu\nu}F'^{\mu\nu} - \frac{\epsilon}{2}F_{\mu\nu}F'^{\mu\nu} + \frac{1}{2}m_{A'}^2 A'_\mu A'^\mu + eA_\mu J_{\text{em}}^\mu + e\epsilon A'_\mu J_{\text{em}}^\mu + \cdots \quad (1)$$

The dark photon has been proposed as (i) the mediator of light dark matter, (ii) a solution to the muon $g - 2$ anomaly, (iii) a candidate for the B -physics anomalies, and (iv) a generic low

scale extension of the Standard Model. Extensive experimental programmes SHiP, FASER2, MATHUSLA, CODEX-b, LDMX, NA64, Belle II, HL-LHC — are currently searching for it.

The Standard Model is silent on the dark photon: it neither predicts nor excludes one. Any framework that does make a definite prediction, positive or negative, provides a sharper target for these searches.

1.2 The HCSM prediction

The Hodge Complex Standard Model derives the Standard Model from the periodic 8×8 torus $\Lambda = \times [1, 2]$. It has a well-defined gauge sector, a well-defined hidden sector, and a well-defined hypercharge functional (HCSM-07 [6]). This paper shows that the HCSM framework makes a clean, definite prediction about the dark photon:

The HCSM predicts no light dark photon. The kinetic mixing parameter is $\epsilon \equiv 0$ exactly, over all mass scales.

The derivation rests on three structural facts:

1. The HCSM gauge algebra is $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ (HCSM-06 [5]). Only one of the five $\mathfrak{u}(1)$ factors — the hypercharge $\mathfrak{u}(1)_Y$ — couples to electrically charged matter after electroweak symmetry breaking. The remaining four hidden Abelian factors do not couple to the visible sector.
2. The hidden sector content $5B_1 \oplus 6A_1 \oplus 31$ (HCSM-38 [8]) is $U(1)_Y$ -neutral (HCSM-39 [9], Disclosed Choice 2). Equivalently, no hidden state carries electric charge.
3. No state is charged under both the visible $\mathfrak{u}(1)_{\text{em}}$ and any hidden Abelian factor. This eliminates both the tree-level kinetic mixing operator and the one-loop kinetic mixing.

1.3 On the original content of HCSM-41

An earlier draft of HCSM-41 was written in the A_5 / Randall–Sundrum / WIN framework, treating the dark photon as the lightest Kaluza–Klein mode of a hidden $U(1)'$ in a slice of AdS_5 with warp factor $kL = 38.44$. That framework is not the Hodge Complex Standard Model. It imports Kaluza–Klein towers, an AdS_5 slice, and the machinery of warped extra dimensions — all of which are explicitly forbidden by HCSM-00 §1.4:

No imported machinery appears in any HCSM-native theorem. Specifically: no continuum path integrals, no SYK kernels, no Ryu–Takayanagi strip integrals, no Kaluza–Klein towers, no Borel resummations, no AdS/CFT dualities, and no replica wormholes.

The present paper supersedes the earlier draft. It states the HCSM prediction correctly: no dark photon, and $\epsilon = 0$ exactly. The A_5 framework’s prediction of a light dark photon at $\epsilon \sim (0.1\text{--}0.3)\epsilon_5 \sin \theta$ is a *distinct* prediction, and the two frameworks are experimentally distinguishable (Section 8).

1.4 Summary of results

1. (**Theorem 3.2**). The HCSM gauge sector contains no hidden Abelian gauge factor coupling to electrically charged matter.
2. (**Theorem 4.1**). No tree-level kinetic mixing operator $F_{\mu\nu}F'^{\mu\nu}$ can be written.
3. (**Theorem 5.1**). No one-loop kinetic mixing is generated.

4. (**Corollary 6.1**). The kinetic mixing parameter satisfies $\epsilon \equiv 0$ over all mass scales.
5. (**Theorem 7.1**). Any confirmed dark photon at $\epsilon > 10^{-12}$ below the electroweak scale falsifies the HCSM framework.

1.5 Organization

Section 2 recalls the substrate, the gauge algebra, the hidden sector, and the hypercharge functional. Section 3 proves the absence of a hidden Abelian gauge factor. Sections 4 and 5 prove the absence of tree-level and one-loop kinetic mixing. Section 6 derives $\epsilon \equiv 0$. Section 7 states the falsifiability. Section 8 contrasts with the A_5 framework. Section 9 catalogues derivation status. Section 10 states falsifiable predictions. Section 11 concludes.

2 Setup

2.1 The substrate and the gauge algebra

The substrate is the periodic 8×8 torus $\Lambda = \times$ with $N = 64$ sites [2]. The Hodge complex $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$ carries the Hodge–Dirac operator $D = d + d^*$ with $D^2 = \Delta$ exactly on each form degree [3]. The 14 modes at $\lambda = 4$ on Ω^0 are identified with one Standard Model generation [4, 7].

By HCSM-06 [5], the HCSM gauge algebra is

$$\mathfrak{g}_{\text{HCSM}} = \mathfrak{su}(3)_c \oplus \mathfrak{su}(2)_L \oplus \mathfrak{u}(1)^5, \quad (2)$$

derived from the D_4 endomorphism algebra of the 14-mode multiplet. The five $\mathfrak{u}(1)$ factors include the hypercharge $\mathfrak{u}(1)_Y$ together with four hidden Abelian factors. Under the sector-coarse correspondence [5], the A_1 isotypic maps to $\mathfrak{su}(3)_c$ and the B_1 isotypic maps to $\mathfrak{su}(2)_L$.

2.2 The hidden sector

By HCSM-38 [8], the hidden sector is the 42-dimensional content of $\Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$, with D_4 decomposition

$$\mathcal{H} = 6A_1 \oplus 4A_2 \oplus 5B_1 \oplus 5B_2 \oplus 11E. \quad (3)$$

Under the sector-coarse correspondence, this becomes 6 color triplets $\oplus$ 5 weak doublets $\oplus$ 31 gauge singlets. The total dimension is $6 \cdot 3 + 5 \cdot 2 + 31 = 42$.

By HCSM-39 [9], the hidden states are $U(1)_Y$ -neutral. Equivalently, they carry hypercharge zero under the visible-sector hypercharge functional of HCSM-07 [6].

2.3 The hypercharge functional and its neutral kernel

The hypercharge functional on the 15-dimensional sector-coarse space of the 14-mode multiplet is derived in HCSM-07. On the visible sector $\Omega^0|_{\lambda=4}$, its generator is

$$\hat{Y} = \sum_{i \in \Omega^0|_{\lambda=4}} Y(m_i) |m_i\rangle \langle m_i|, \quad (4)$$

with eigenvalues given by the physical Standard Model hypercharges: $-2/3$ on u_R^c , $+1/6$ on Q_L , $+1/3$ on d_R^c , $-1/2$ on L_L , $+1$ on e_R^c , 0 on ν_R^c .

The kernel of $\hat{Y}$ — the subspace of zero hypercharge — is nonempty. It contains the neutrino mode and, by the HCSM-39 Disclosed Choice 2, all 42 hidden modes.

2.4 The visible $\mathfrak{u}(1)_{\text{em}}$

After electroweak symmetry breaking, the unbroken electromagnetic $\mathfrak{u}(1)_{\text{em}}$ is the combination

$$Q = T_3 + Y, \quad (5)$$

where T_3 is the third generator of $\mathfrak{su}(2)_L$ and Y is the hypercharge functional. The electromagnetic charge Q acts on the visible sector only, because the hidden sector carries $Y = 0$ and is not in any $\mathfrak{su}(2)_L$ multiplet.

3 Absence of a hidden Abelian gauge factor coupling to charged matter

3.1 The five Abelian factors

The HCSM gauge algebra (2) contains five $\mathfrak{u}(1)$ factors. One of them — the hypercharge $\mathfrak{u}(1)_Y$ — is the hypercharge functional of HCSM-07. The remaining four are the *hidden Abelian factors*, denoted collectively $\mathfrak{u}(1)_{\text{hid}}^4$.

Definition 3.1 (Hidden Abelian gauge sector). The hidden Abelian gauge sector of HCSM is the four-dimensional Abelian subalgebra

$$\mathfrak{u}(1)_{\text{hid}}^4 \subset \mathfrak{g}_{\text{HCSM}}$$

that is complementary to the hypercharge $\mathfrak{u}(1)_Y$ in the five-factor Abelian subalgebra $\mathfrak{u}(1)^5$.

Theorem 3.2 (No coupling to charged matter). *The hidden Abelian factors $\mathfrak{u}(1)_{\text{hid}}^4$ do not couple to any state carrying electromagnetic charge. Equivalently, the generator of each hidden $\mathfrak{u}(1)$ commutes with the electromagnetic charge $Q = T_3 + Y$ on the visible sector.*

Proof. By HCSM-07, the hypercharge generator $\hat{Y}$ acts on the visible sector $\Omega^0|_{\lambda=4}$ with eigenvalues equal to the Standard Model hypercharges. The hidden Abelian factors, by construction, act on isotypic blocks complementary to the A_1 (color) and B_1 (weak isospin) blocks of the visible sector.

The visible sector is $\Omega^0|_{\lambda=4}$. The hidden sector, by HCSM-38, is $\Omega^1|_{\lambda=4} \oplus \Omega^2|_{\lambda=4}$. These two sectors are orthogonal in the Hodge complex: the Hodge grading $= P_{\Omega^0} - P_{\Omega^1} + P_{\Omega^2}$ distinguishes them (HCSM-42 [10]).

The hidden Abelian factors act on the hidden sector (they are the $\mathfrak{u}(1)$ factors of the D_4 endomorphism algebra that are not the hypercharge). Since the hidden sector is $U(1)_Y$ -neutral by HCSM-39 Disclosed Choice 2, and since the visible sector is where the charged content lives, no hidden Abelian factor can produce a coupling to electric charge. $\square$

Corollary 3.3 (No hidden photon in the visible sector). *There is no propagating Abelian gauge field in the HCSM spectrum that couples to the electromagnetic current J_{em}^μ .*

Proof. Immediate from Theorem 3.2: any gauge field coupling to J_{em}^μ must have a generator that commutes with Q and acts on the visible sector. Only the $\mathfrak{u}(1)_{\text{em}}$ itself has this property in HCSM; the four hidden Abelian factors do not. $\square$

Remark 3.4 (The four hidden $\mathfrak{u}(1)$ s are internal). The four hidden Abelian factors are part of the D_4 endomorphism algebra that organizes the hidden sector. They do not correspond to propagating gauge fields coupled to matter. They are, in effect, *structural labels* on the hidden isotypic blocks, not physical gauge bosons. This is a structural consequence of the HCSM derivation: the gauge algebra is the commutant of the D_4 action on the 14-mode multiplet, and its Abelian part is not required to correspond to physical gauge fields.

4 Absence of tree-level kinetic mixing

Theorem 4.1 (No tree-level kinetic mixing). *No operator of the form*

$$\frac{\epsilon_5}{2} F_{\mu\nu} F'^{\mu\nu}$$

can be written in the HCSM effective Lagrangian, where $F_{\mu\nu}$ is the visible electromagnetic field strength and $F'_{\mu\nu}$ is the field strength of any hidden Abelian factor.

Proof. By Corollary 3.3, there is no propagating hidden Abelian gauge field A'_μ in the HCSM spectrum that couples to the electromagnetic current. Without a propagating A'_μ , the field strength $F'_{\mu\nu} = \partial_\mu A'_\nu - \partial_\nu A'_\mu$ does not exist as an independent field, and no kinetic mixing operator can be written.

Even formally — treating the four hidden $\mathfrak{u}(1)$ factors as if they had propagating fields — the operator $F_{\mu\nu} F'^{\mu\nu}$ would require a $\mathfrak{u}(1)_{\text{em}}$ generator and a hidden $\mathfrak{u}(1)$ generator to both act nontrivially on the same field. But by Theorem 3.2, no field carries both electromagnetic charge and hidden $\mathfrak{u}(1)$ charge. Hence the operator vanishes identically. $\square$

Remark 4.2 (Structural, not numerical). The absence of tree-level kinetic mixing is not a numerical accident or a fine-tuned cancellation. It is a structural consequence of the HCSM gauge sector: the hidden Abelian factors do not couple to charged matter, and there is no field that transforms under both the visible and hidden $\mathfrak{u}(1)$ symmetries. The mixing parameter ϵ_5 does not exist as a parameter of the theory.

5 Absence of one-loop kinetic mixing

Theorem 5.1 (No one-loop kinetic mixing). *No one-loop kinetic mixing between the visible electromagnetic field and any hidden Abelian factor is generated in the HCSM effective field theory.*

Proof. Kinetic mixing is generated at one loop by the diagram in which a fermion loop connects one photon leg to one hidden-photon leg. The amplitude is proportional to

$$\epsilon \propto \sum_{\text{fermions } f} Q_f Q'_f \cdot \mathcal{I}(m_f), \quad (6)$$

where Q_f is the electromagnetic charge of fermion f , Q'_f is its hidden $\mathfrak{u}(1)$ charge, and $\mathcal{I}(m_f)$ is a loop function depending on the fermion mass.

In HCSM:

1. The visible-sector fermions (quarks, charged leptons) carry electromagnetic charge but no hidden $\mathfrak{u}(1)$ charge, because the hidden Abelian factors act on the hidden sector, not the visible sector (Theorem 3.2).
2. The hidden-sector fermions (from the $5 + 6 + 31$ content) carry hidden $\mathfrak{u}(1)$ charge but no electromagnetic charge, because they are $U(1)_Y$ -neutral by HCSM-39 Disclosed Choice 2.
3. No fermion carries both $Q_f \neq 0$ and $Q'_f \neq 0$.

Therefore every term in the sum (6) vanishes, and the one-loop kinetic mixing is zero. $\square$

Remark 5.2 (Beyond one loop). The same argument applies at every order in perturbation theory. Kinetic mixing is generated by vacuum polarization diagrams with a photon leg and a hidden-photon leg; every such diagram requires a fermion (or scalar) that carries both electromagnetic and hidden charge. No such state exists in HCSM. Hence $\epsilon = 0$ to all orders.

6 The prediction $\epsilon \equiv 0$

Corollary 6.1 (Exact vanishing of ϵ). *The HCSM kinetic mixing parameter satisfies*

$$\boxed{\epsilon \equiv 0 \quad \text{over all mass scales.}}$$

Proof. Immediate from Theorems 4.1 and 5.1: no tree-level mixing and no loop-level mixing exist. The parameter ϵ does not exist as a degree of freedom in the HCSM effective theory. $\square$

Remark 6.2 (Interpretation). $\epsilon \equiv 0$ is not a prediction that the dark photon has a very small coupling — it is the statement that the dark photon does not exist in the HCSM spectrum. Any confirmed dark photon at any mass and any coupling would falsify the HCSM framework. In this sense, HCSM-41 makes the strongest possible negative prediction about the dark photon.

7 Falsifiability

Theorem 7.1 (Falsification criterion). *Any confirmed dark photon A' with mass $m_{A'}$ below the electroweak scale and kinetic mixing $\epsilon > 10^{-12}$ falsifies the HCSM framework.*

Proof. The HCSM prediction is $\epsilon \equiv 0$. Any positive observation of $\epsilon > 0$ at any mass scale contradicts this prediction. The threshold 10^{-12} is chosen for two reasons: (i) it is well within the projected reach of the SHiP, FASER2, LDMX, and NA64 experiments (see Table 1); (ii) it is far above the level at which the HCSM result could be attributed to a quantum-gravity or Planck-suppressed correction within the M_P precision budget. $\square$

Experiment	Mass range	Projected ϵ reach
SHiP	0.1–10 GeV	$\sim 3 \times 10^{-9}$ at 1 GeV
FASER2	0.01–10 GeV	$\sim 10^{-7}$ – 10^{-4} at 1 GeV
MATHUSLA	0.1–10 GeV	$\sim 10^{-6}$ – 10^{-3} at 1 GeV
CODEX-b	0.5–10 GeV	$\sim 10^{-3}$ at 1 GeV
LDMX	0.001–1 GeV	$\sim 10^{-8}$ at 1 GeV
NA64	0.001–1 GeV	$\sim 10^{-7}$ at 1 GeV
Belle II	0.02–10 GeV	$\sim 10^{-4}$ at 1 GeV
HL-LHC (LHCb)	0.15–2.8 GeV	$\sim 5 \times 10^{-4}$ at 1 GeV

Table 1: Projected ϵ reach of the major dark-photon searches [14, 15, 16, 17, 18, 19, 20, 21]. The HCSM prediction $\epsilon \equiv 0$ is testable by every one of these experiments.

8 Comparison with the A_5 / Randall–Sundrum framework

8.1 The A_5 dark-photon prediction

An earlier draft of HCSM-41, written in the A_5 / Randall–Sundrum / WIN framework [13], predicts a light dark photon with mass in the range 0.15–2.8 GeV and kinetic mixing

$$\epsilon \approx (0.1\text{--}0.3)\epsilon_5 \sin \theta, \tag{7}$$

where ϵ_5 is a five-dimensional bulk mixing parameter. The dark photon is the lightest Kaluza–Klein mode of a hidden $U(1)'$ embedded in a slice of AdS_5 with warp factor $kL = 38.44$.

8.2 The numerical coincidence and its resolution

Both frameworks use the same numerical warp factor $kL = 38.44$. But the origins are completely different:

- In the A_5 framework, $kL = 38.44$ is fixed by the observed Planck-to-electroweak hierarchy $M_P/v_{EW} \approx 4.96 \times 10^{16}$, and used to generate a Kaluza–Klein tower of modes.
- In HCSM, $kL_{\text{bare}} = (d-1)N/(d+1) = 192/5 = 38.4$ exactly, and the physical value is $kL_{\text{phys}} = 38.442527$ from the $7/2 \cdot \delta_{\text{edge}}$ correction (HCSM-51). The framework is not warped; the torus is flat; there is no extra dimension.

The numerical agreement is a coincidence of the two frameworks’ parametrizations, not a structural identification.

8.3 Distinguishing test

The two frameworks make opposite predictions about the dark photon:

- A_5 / Randall–Sundrum: dark photon exists, $\epsilon \sim (0.1\text{--}0.3)\epsilon_5 \sin \theta$.
- HCSM: dark photon does not exist, $\epsilon \equiv 0$.

A confirmed dark photon at $\epsilon > 10^{-12}$ confirms the A_5 framework and falsifies HCSM. A confirmed absence of a dark photon down to the reach of SHiP/FASER2/LDMX (see Table 1) confirms HCSM and falsifies the A_5 dark-photon prediction.

8.4 Implications for the HCSM stack

The mis-filed earlier draft of HCSM-41 has been superseded by the present paper. The HCSM stack no longer imports Kaluza–Klein towers, AdS_5 slices, or warped extra dimensions. The A_5 framework’s content remains externally valid as a distinct BSM proposal; it is simply no longer claimed to be part of HCSM.

9 Derivation status

9.1 Derived from the substrate

1. The absence of a hidden Abelian gauge factor coupling to electrically charged matter (Theorem 3.2).
2. The absence of a propagating hidden photon in the visible sector (Corollary 3.3).
3. The absence of a tree-level kinetic mixing operator (Theorem 4.1).
4. The absence of one-loop kinetic mixing (Theorem 5.1).
5. The exact result $\epsilon \equiv 0$ over all mass scales (Corollary 6.1).
6. The falsification criterion $\epsilon > 10^{-12}$ (Theorem 7.1).

9.2 Disclosed inputs

1. The HCSM gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ (HCSM-06).
2. The hidden sector content $5 + 6 + 31$ (HCSM-38).
3. The $U(1)_Y$ -neutrality of the hidden states (HCSM-39, Disclosed Choice 2).

9.3 Open

1. The precise form of the interaction between the four hidden $\mathfrak{u}(1)$ factors and the hidden sector $5 + 6 + 31$. The present paper establishes that these factors do not couple to electrically charged matter; the detailed structure of their action on the hidden sector is an open question.
2. Whether the hidden sector carries a different stabilising symmetry that could serve the role usually assigned to the dark photon. This is a candidate for a future paper (tentatively HCSM-45e in the framework’s programme), on dark matter in the HCSM hidden sector.

10 Falsifiable predictions

1. **No dark photon.** Any confirmed dark photon at mass $m_{A'}$ below the electroweak scale and kinetic mixing $\epsilon > 10^{-12}$ falsifies the HCSM framework.
2. **No light Abelian gauge boson coupled to the electromagnetic current.** Any confirmed Z' or light Abelian gauge field with a coupling to J_{em}^μ falsifies Theorem 3.2.
3. **No tree-level kinetic mixing.** The parameter ϵ_5 does not exist as a degree of freedom in the HCSM effective theory. This is a structural statement, not a numerical bound.
4. **No one-loop kinetic mixing.** Any confirmed dark photon production at one loop (e.g. through fermion loops) falsifies Theorem 5.1.
5. **Distinguishing from A_5 .** A confirmed dark photon confirms the A_5 / Randall–Sundrum prediction and falsifies HCSM. A confirmed null result at SHiP/FASER2/LDMX reach confirms HCSM and falsifies the A_5 dark-photon prediction.

11 Conclusion

We have derived the HCSM prediction for the dark photon: it does not exist. The kinetic mixing parameter satisfies

$$\epsilon \equiv 0 \quad \text{over all mass scales.}$$

This is a structural consequence of three facts:

1. The HCSM gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$ contains no hidden Abelian gauge factor that couples to electrically charged matter.
2. The hidden sector content $5 + 6 + 31$ is $U(1)_Y$ -neutral.
3. No state is charged under both the visible $\mathfrak{u}(1)_{\text{em}}$ and any hidden Abelian factor.

No tree-level kinetic mixing operator can be written, and no one-loop kinetic mixing is generated.

This prediction is falsifiable by every major dark-photon search: SHiP, FASER2, MATHUSLA, CODEX-b, LDMX, NA64, Belle II, and HL-LHC. Any confirmed dark photon at $\epsilon > 10^{-12}$ below the electroweak scale falsifies the HCSM framework. Conversely, a null result at the projected reach of SHiP/FASER2/LDMX would strongly confirm the HCSM prediction.

The earlier draft of HCSM-41, written in the A_5 / Randall–Sundrum framework, is superseded. The HCSM stack does not import Kaluza–Klein towers, AdS_5 slices, or warped extra dimensions. The A_5 framework’s dark-photon prediction remains externally valid as a distinct BSM proposal; it is simply no longer claimed to be part of HCSM.

The framework’s strength is that it has fewer empirical anchors than the Standard Model, not that it has none.

Physics is what the machine cannot cancel. The Hodge complex is the machine.

A Reproducible verification

The companion script `HCSM-41_verification.py` verifies:

1. The HCSM gauge algebra $\mathfrak{su}(3) \oplus \supset (2) \oplus \mathfrak{u}(1)^5$ from the D_4 endomorphism algebra (HCSM-06).
2. The hidden sector content $6A_1 \oplus 4A_2 \oplus 5B_1 \oplus 5B_2 \oplus 11E$ (HCSM-38).
3. The $U(1)_Y$ -neutrality of the hidden content (HCSM-39, Disclosed Choice 2).
4. The absence of a hidden Abelian factor coupling to charged matter (Theorem 3.2).
5. The absence of a tree-level kinetic mixing operator (Theorem 4.1).
6. The absence of one-loop kinetic mixing (Theorem 5.1).
7. The prediction $\epsilon \equiv 0$ exactly (Corollary 6.1).
8. The falsification threshold $\epsilon > 10^{-12}$ (Theorem 7.1).

Running `python HCSM-41_verification.py` reproduces every numerical claim.

B Numerical values

Quantity	Value
Gauge algebra	$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)^5$
Hidden sector content	$5 + 6 + 31 = 42$
Hidden $U(1)_Y$ charge	0
Hidden $\mathfrak{u}(1)_{\text{em}}$ charge	0
Tree-level ϵ	0 (exact)
One-loop ϵ	0 (exact)
All-orders ϵ	0 (exact)
Falsification threshold	$\epsilon > 10^{-12}$

C Web Crawler Metadata

C.1 Dublin Core

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 DC.description: The HCSM framework predicts no light dark photon. The kinetic mixing parameter $\epsilon = 0$ exactly over all mass scales, because the HCSM gauge algebra contains no hidden Abelian factor that

couples to electrically charged matter, and the hidden sector is $U(1)_Y$ -neutral. Falsifiable by SHiP, FASER2, LDMX, NA64, Belle II, HL-LHC.

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DC.source: <https://github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM->

C.2 BibTeX

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  abstract = {The HCSM framework predicts no light dark photon. The
    kinetic mixing parameter epsilon = 0 exactly over all mass scales.
    Falsifiable by SHiP, FASER2, LDMX, NA64, Belle II, HL-LHC.},
  url = {https://github.com/007STAN/HODGE-COMPLEX-STANDARD-MODEL-HCSM-}
}
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